

# EE227 – Digital Logic Design

Lecture 19 – 20

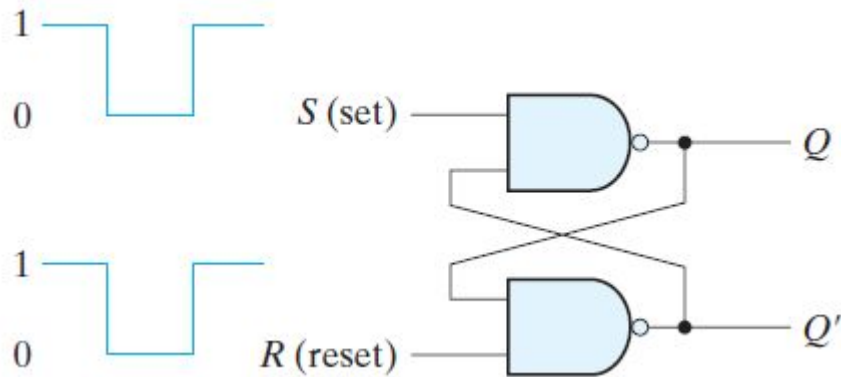
# Outline

- Flip Flops
- D – Flip Flop
- JK Flip Flop
- T – Flip Flop
- Quiz 3 Solution
- Code Conversion Combinational Circuit
- Quiz 4
- Quiz 4 Solution

# Latch and Flip Flop

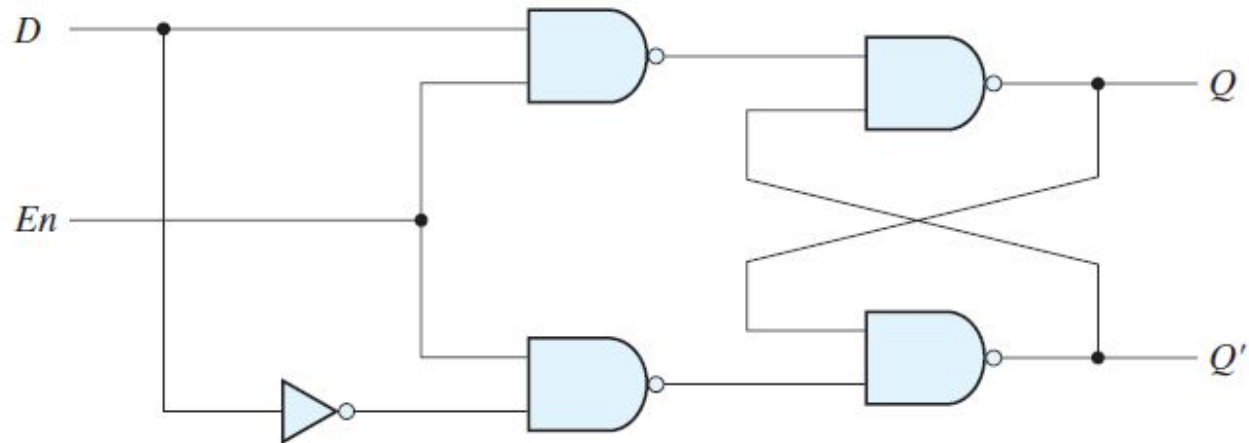
- Latches
  - Storage elements that operate with signal levels are referred to as latches
- Flip Flops
  - Storage elements that are controlled by a clock transition are flip-flops

# SR Latch



| <b>S</b> | <b>R</b> | <b>Q</b>       | <b>State</b> |
|----------|----------|----------------|--------------|
| 0        | 0        | ?              | Forbidden    |
| 0        | 1        | 1              | Set          |
| 1        | 0        | 0              | Reset        |
| 1        | 1        | Previous State | No Change    |

# D Latch



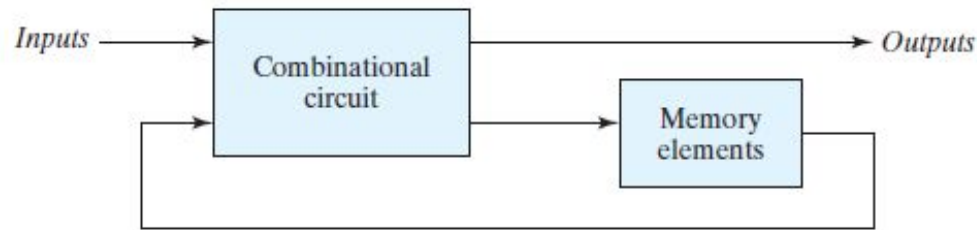
| En | D | Next State of Q                 |
|----|---|---------------------------------|
| 0  | X | No Change                       |
| 1  | 0 | $Q = 0 \rightarrow$ Reset State |
| 1  | 1 | $Q = 1 \rightarrow$ Set State   |

# Conversion of D Latch to Flip Flop

- The state of a latch or flip-flop is switched by a change in the control input
- This momentary change is called a trigger, and the transition it causes is said to trigger the flip-flop
- The D latch with pulses in its control input is essentially a flip-flop that is triggered every time the pulse goes to the logic-1 level
- As long as the pulse input remains at this level, any changes in the data input will change the output and the state of the latch

# Problem with using Latches in Sequential Circuits (1/2)

- In a sequential circuit the inputs of the flip-flops are derived in part from the outputs of the same and other flip-flops



- The state transitions of the latches start as soon as the clock pulse changes to the logic-1 level
- The new state of a latch appears at the output while the pulse is still active
- If the inputs applied to the latches change while the clock pulse is still at the logic-1 level, the latches will respond to new values and a new output state may occur
- The result is an unpredictable situation

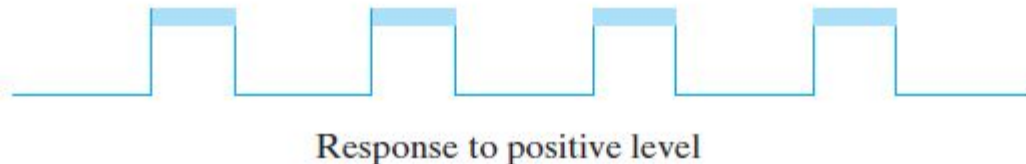
# Problem with using Latches in Sequential Circuits (2/2)

- If the inputs applied to the latches change while the clock pulse is still at the logic-1 level, the latches will respond to new values and a new output state may occur
- The result is an unpredictable situation
- Because of this unreliable operation, the output of a latch cannot be applied directly or through combinational logic to the input of the same or another latch when all the latches are triggered by a common clock source



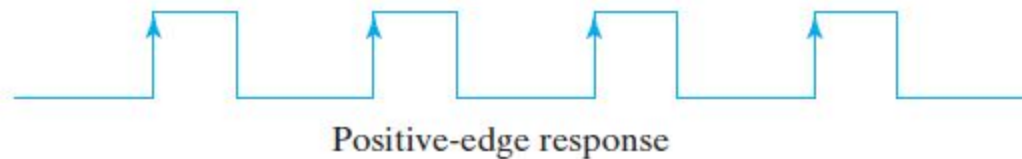
# Flip Flops (1/3)

- Flip-flop circuits are constructed in such a way as to make them operate properly when they are part of a sequential circuit that employs a common clock
- The problem with the latch is that it responds to a change in the level of a clock pulse

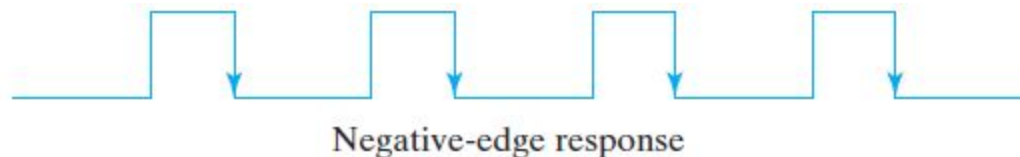


# Flip Flops (2/3)

- The key to the proper operation of a flip-flop is to trigger it only during a signal transition
- A clock pulse goes through two transitions:
  - From 0 to 1
    - Positive edge transition



- From 1 to 0
  - Negative edge transition

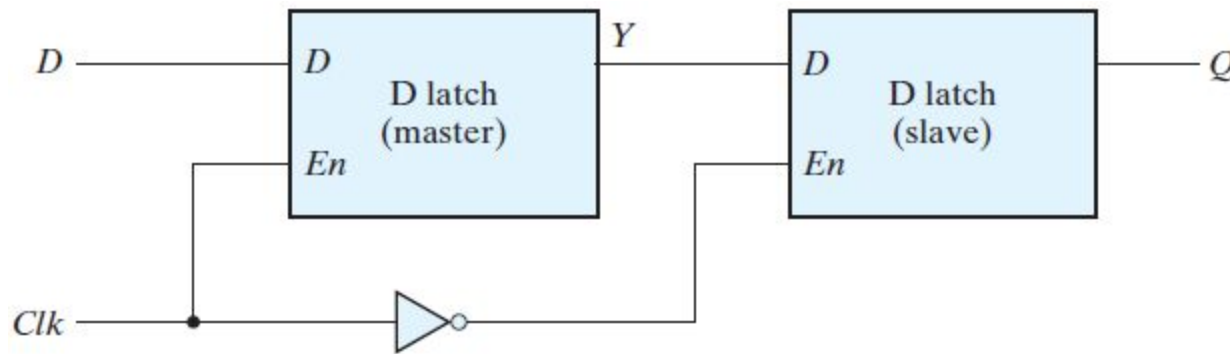


# Flip Flops (3/3)

- There are two ways that a latch can be modified to form a flip-flop
  - One way is to employ two latches in a special configuration that isolates the output of the flip-flop and prevents it from being affected while the input to the flip-flop is changing
  - Another way is to produce a flip-flop that triggers only during a signal transition (from 0 to 1 or from 1 to 0) of the synchronizing signal (clock) and is disabled during the rest of the clock pulse

# Edge Triggered D – Flip Flop with D Latches

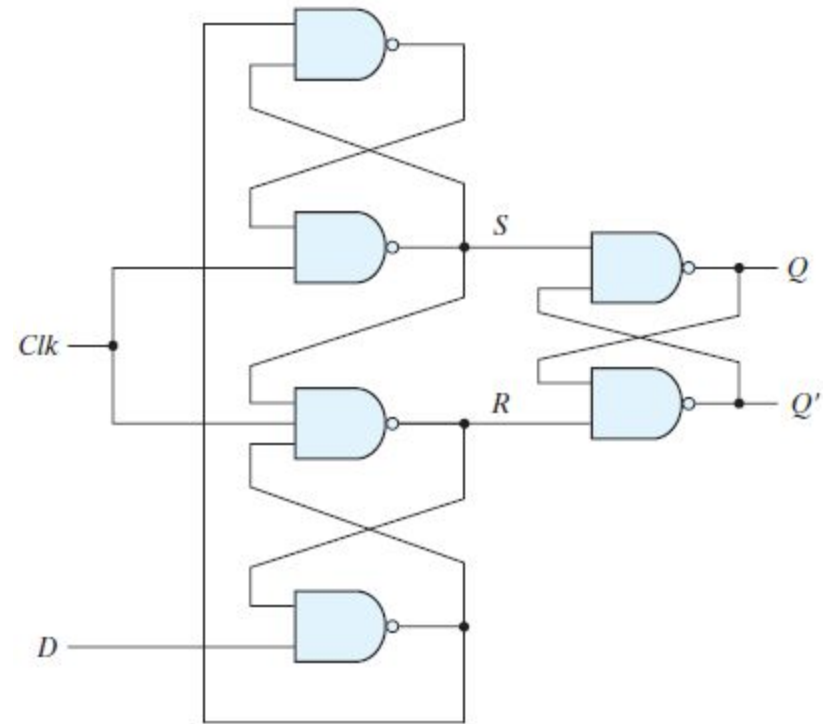
- Master – Slave Configuration
- A change in the output of the flip-flop can be triggered only by and during the transition of the clock from 1 to 0



| $D$ | $Q(t + 1)$ |       |
|-----|------------|-------|
| 0   | 0          | Reset |
| 1   | 1          | Set   |

# Edge Triggered D – Flip Flop with SR Latches

- When  $\text{Clk} = 0$ 
  - $R = 1, S = 1$
  - No change in state
- When  $D = 0$  and  $\text{Clk}$  changes from 0 to 1
  - $R = 0, S = 1 \Rightarrow Q = 0$
  - Reset state
  - Further change in  $D$  when  $\text{Clk} = 1$
  - Locked state as  $R = 0$
- When  $D = 1$  and  $\text{Clk}$  changes from 0 to 1
  - $S = 0, R = 1 \Rightarrow Q = 1$
  - Set State

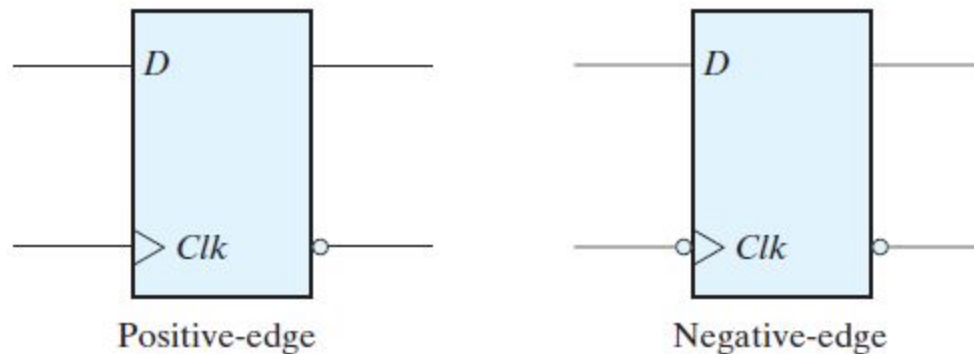


# Setup, Hold and Propagation Delay Time

- There is a minimum time called the **setup time** during which the D input must be maintained at a constant value prior to the occurrence of the clock transition
- There is a minimum time called the **hold time** during which the D input must not change after the application of the positive transition of the clock
- The **propagation delay time** of the flip-flop is defined as the interval between the trigger edge and the stabilization of the output to a new state

# Graphical Symbols of D – Flip Flop

- The dynamic indicator (>) denotes the fact that the flip-flop responds to the edge transition of the clock



# Other Flip Flops

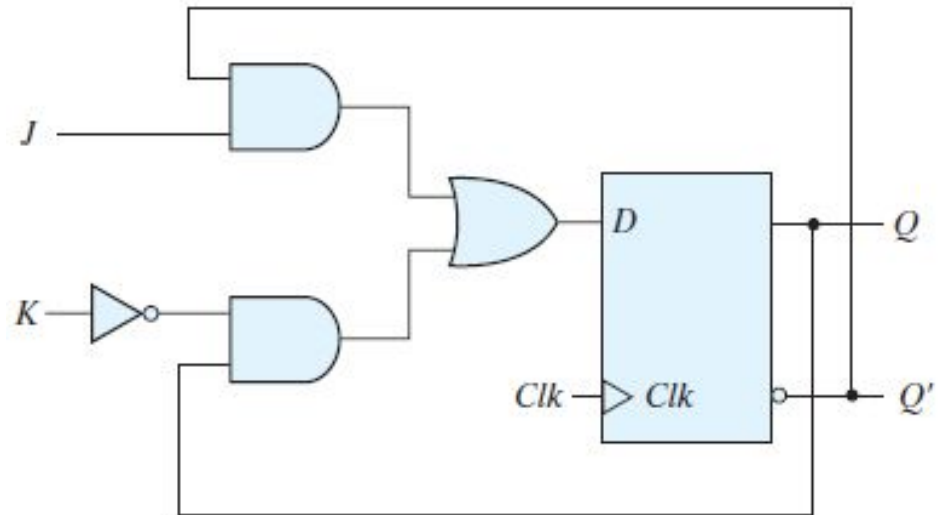
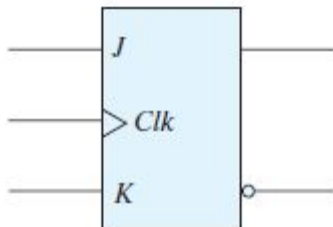
- There are three operations that can be performed with a flip-flop
  - Set it to 1
  - Reset it to 0
  - Complement its output
- With only a single input, the D flip-flop can set or reset the output
- There are two other flip flops
  - JK flip flop
  - T flip flop



# JK Flip Flop

- 2 inputs J and K
- Performs all 3 operations
- $D = JQ' + K'Q$

| <i>J</i> | <i>K</i> | $Q(t + 1)$ |            |
|----------|----------|------------|------------|
| 0        | 0        | $Q(t)$     | No change  |
| 0        | 1        | 0          | Reset      |
| 1        | 0        | 1          | Set        |
| 1        | 1        | $Q'(t)$    | Complement |

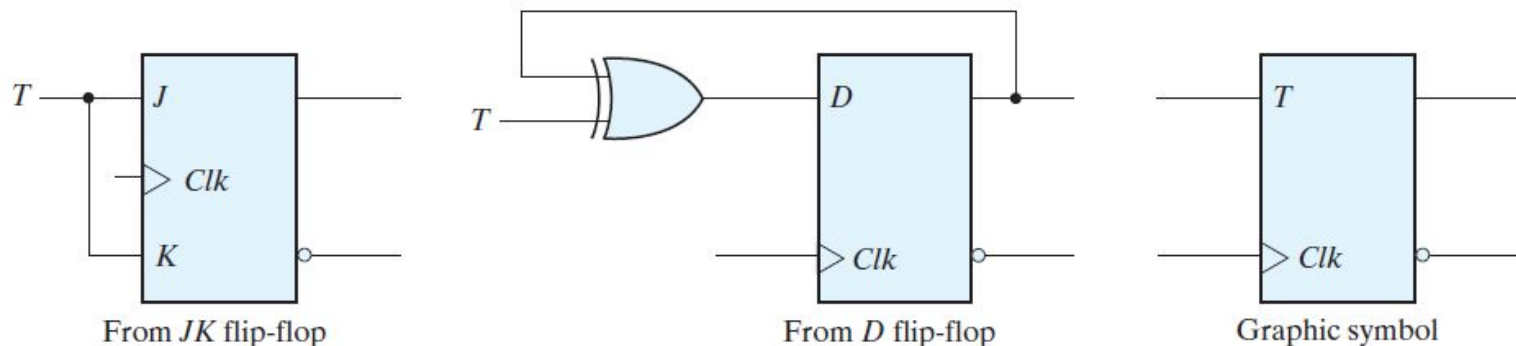


# Toggle (T) – Flip Flop

- The T (toggle) flip-flop is a complementing flip-flop and can be obtained from a JK flip-flop when inputs J and K are tied together
- The T flip-flop can be constructed with a D flip-flop and an exclusive-OR gate as

$$D = T \oplus Q = TQ' + T'Q$$

| $T$ | $Q(t + 1)$ |            |
|-----|------------|------------|
| 0   | $Q(t)$     | No change  |
| 1   | $Q'(t)$    | Complement |



# Characteristic Equations

- The next state  $[Q(t + 1)]$  can be obtained with the help of characteristic equations

- For D – Flip Flop

$$Q(t + 1) = D$$

- For JK Flip Flop

$$Q(t + 1) = JQ' + K'Q$$

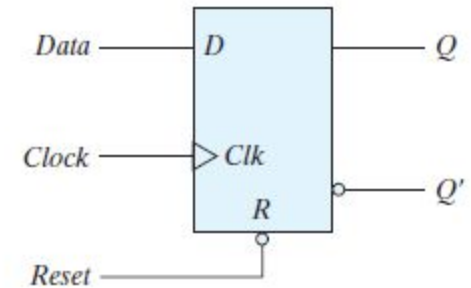
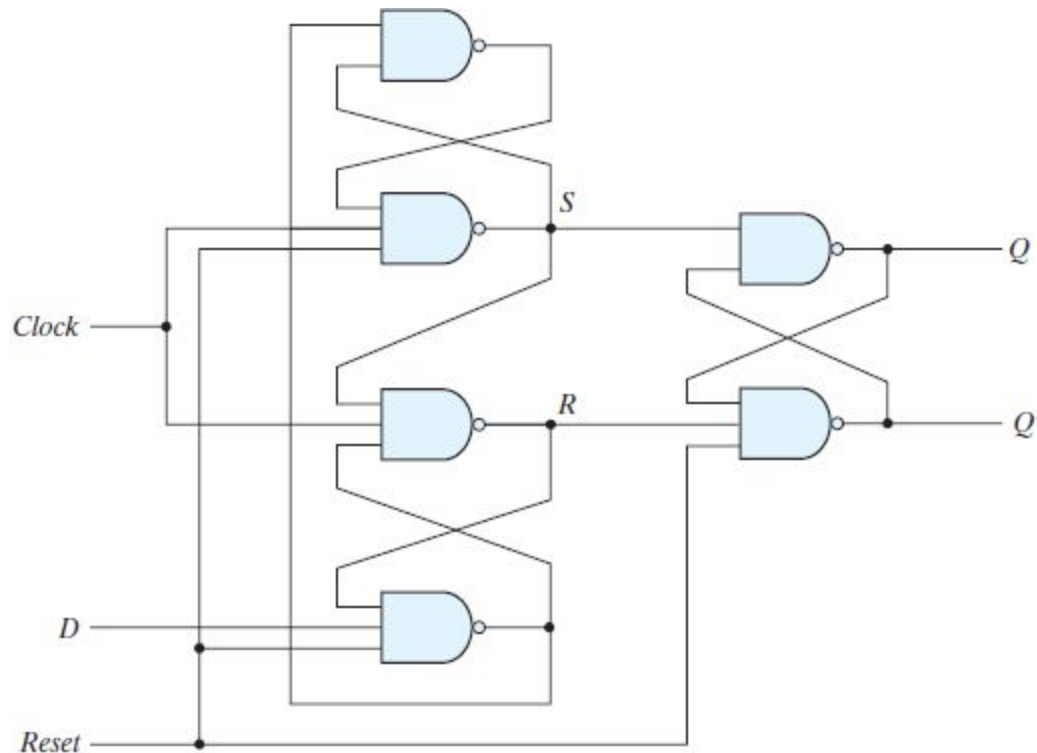
- For T – Flip Flop

$$Q(t + 1) = T \oplus Q = TQ' + T'Q$$

# Direct Inputs in Flip Flops

- Some flip-flops have asynchronous inputs that are used to force the flip-flop to a particular state independently of the clock
- The input that sets the flip-flop to 1 is called **preset** or **direct set**
- The input that clears the flip-flop to 0 is called **clear** or **direct reset**

# Edge Triggered D Flip Flop with Direct Reset



| <i>R</i> | <i>Clk</i> | <i>D</i> | <i>Q</i> | <i>Q'</i> |
|----------|------------|----------|----------|-----------|
| 0        | X          | X        | 0        | 1         |
| 1        | ↑          | 0        | 0        | 1         |
| 1        | ↑          | 1        | 1        | 0         |

# Code Conversion Combinational Circuit

Missed Topic from Chapter 4

# BCD to Excess – 3 Code Conversion (1/4)

- Truth Table

| Inputs |   |   |   | Outputs |   |   |   |
|--------|---|---|---|---------|---|---|---|
| A      | B | C | D | w       | x | y | z |
| 0      | 0 | 0 | 0 | 0       | 0 | 1 | 1 |
| 0      | 0 | 0 | 1 | 0       | 1 | 0 | 0 |
| 0      | 0 | 1 | 0 | 0       | 1 | 0 | 1 |
| 0      | 0 | 1 | 1 | 0       | 1 | 1 | 0 |
| 0      | 1 | 0 | 0 | 0       | 1 | 1 | 1 |
| 0      | 1 | 0 | 1 | 1       | 0 | 0 | 0 |
| 0      | 1 | 1 | 0 | 1       | 0 | 0 | 1 |
| 0      | 1 | 1 | 1 | 1       | 0 | 1 | 0 |
| 1      | 0 | 0 | 0 | 1       | 0 | 1 | 1 |
| 1      | 0 | 0 | 1 | 1       | 1 | 0 | 0 |
| 1      | 0 | 1 | 0 | X       | X | X | X |
| 1      | 0 | 1 | 1 | X       | X | X | X |
| 1      | 1 | 0 | 0 | X       | X | X | X |
| 1      | 1 | 0 | 1 | X       | X | X | X |
| 1      | 1 | 1 | 0 | X       | X | X | X |
| 1      | 1 | 1 | 1 | X       | X | X | X |

# BCD to Excess – 3 Code Conversion (2/4)

- K – Map Simplification

|    |    | CD            |               |               |               |
|----|----|---------------|---------------|---------------|---------------|
|    |    | 00            | 01            | 11            | 10            |
| AB | 00 | $m_0$<br>1    | $m_1$         | $m_3$         | $m_2$<br>1    |
|    | 01 | $m_4$<br>1    | $m_5$         | $m_7$         | $m_6$<br>1    |
|    | 11 | $m_{12}$<br>X | $m_{13}$<br>X | $m_{15}$<br>X | $m_{14}$<br>X |
|    | 10 | $m_8$<br>1    | $m_9$         | $m_{11}$<br>X | $m_{10}$<br>X |

$$z = D'$$

|    |    | CD            |               |               |               |
|----|----|---------------|---------------|---------------|---------------|
|    |    | 00            | 01            | 11            | 10            |
| AB | 00 | $m_0$<br>1    | $m_1$         | $m_3$<br>1    | $m_2$         |
|    | 01 | $m_4$<br>1    | $m_5$         | $m_7$<br>1    | $m_6$         |
|    | 11 | $m_{12}$<br>X | $m_{13}$<br>X | $m_{15}$<br>X | $m_{14}$<br>X |
|    | 10 | $m_8$<br>1    | $m_9$         | $m_{11}$<br>X | $m_{10}$<br>X |

$$y = CD + C'D'$$

|    |    | CD            |               |               |               |
|----|----|---------------|---------------|---------------|---------------|
|    |    | 00            | 01            | 11            | 10            |
| AB | 00 | $m_0$         | $m_1$<br>1    | $m_3$<br>1    | $m_2$<br>1    |
|    | 01 | $m_4$<br>1    | $m_5$         | $m_7$         | $m_6$         |
|    | 11 | $m_{12}$<br>X | $m_{13}$<br>X | $m_{15}$<br>X | $m_{14}$<br>X |
|    | 10 | $m_8$         | $m_9$<br>1    | $m_{11}$<br>X | $m_{10}$<br>X |

$$x = B'C + B'D + BC'D'$$

|    |    | CD            |               |               |               |
|----|----|---------------|---------------|---------------|---------------|
|    |    | 00            | 01            | 11            | 10            |
| AB | 00 | $m_0$         | $m_1$         | $m_3$         | $m_2$         |
|    | 01 | $m_4$         | $m_5$<br>1    | $m_7$<br>1    | $m_6$<br>1    |
|    | 11 | $m_{12}$<br>X | $m_{13}$<br>X | $m_{15}$<br>X | $m_{14}$<br>X |
|    | 10 | $m_8$<br>1    | $m_9$<br>1    | $m_{11}$<br>X | $m_{10}$<br>X |

$$w = A + BC + BD$$



# BCD to Excess – 3 Code Conversion (3/4)

- Boolean Equations

$$z = D'$$

$$y = CD + C'D'$$

$$y = CD + (C + D)'$$

$$x = B'C + B'D + BC'D'$$

$$x = B'(C + D) + BC'D'$$

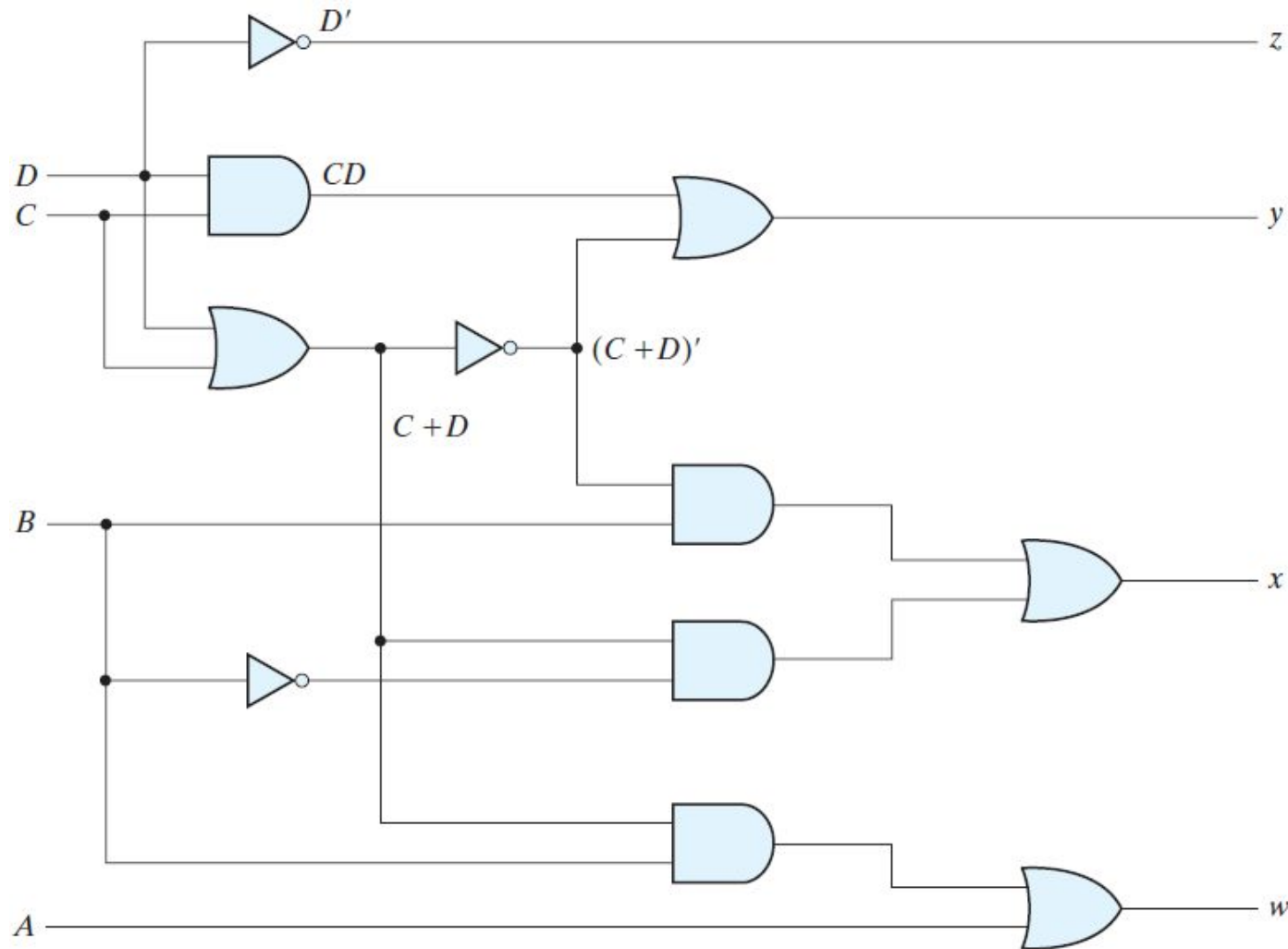
$$x = B'(C + D) + B(C + D)'$$

$$w = A + BC + BD$$

$$w = A + B(C + D)$$

# BCD to Excess – 3 Code Conversion (4/4)

- Circuit Diagram



# Quiz 3 and 4

Solution

Was Done in Class